Subject

41091 - Data Systems

6 Credit points

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- Subject description
- Availabilities
- Learning outcomes
- Learning and teaching activities
- Assessment information
- Requisites
- Additional subject details

Subject description

This subject delves into systems engineering for complex, data-centric systems, covering their lifecycle from design to management. You'll learn to build data pipelines through Extract, Transform, Load processes, apply Python for data engineering and reporting, and explore the Hadoop ecosystem and cloud data system providers. The learning experience is project-based, combining workshops on data engineering with collaborative project work, equipping you with practical, in-demand skills.

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Faculty
Engineering and Information Technology

Subject type
Subject

Study level
Undergraduate

Result type
Grade and marks

Total Workload hours
150

Apply now

Important dates

Fee calculator

Availabilities

Click on Select availability in the top right of screen and choose from the location, attendance mode and session options.

Learning outcomes

Collapse all

Upon completion of this subject, graduates will be able to:

1. Define requirements and specifications of data system. (B.1)
2. Demonstrate a systematic design and thinking process for data system development. (C.1)
3. Prototype or implement data system according to system specifications. (D.1)
4. Communicate effectively to data system project stakeholders. (E.1)
5. Work effectively in a team to develop a data system, demonstrating continuous improvement and independence. (E.1, F.1)

Learning and teaching activities

The subject is carried out through the development of data systems in collaborative classes.

The collaborative classes are designed to support and guide the students through the development life cycle in data system project. The class is composed of workshop session and project development session. In the workshop session, theories and concepts of data system development will be discussed. In the project development session, students will participate in team work to develop a data system.

At the beginning of the semester, each student will join a team. Each team will choose a data system project. This project will be carried out through the development phases of planning, requirements and specification, design, implementation and testing. The students will present their project work in the class. Other teams are required to review and critique other team's presentations.

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Assessment information

Click on Select availability in the top right of screen and choose from the location, attendance mode and session options.

Requisites

Requisite detail

Before you enrol in this subject, you must ensure you have met its requisites. The requisites for this subject can be viewed on its access conditions page.

To learn more about requisites, please visit the How to use the Handbook guide.

Additional subject details

Collapse all

Professional standards

Engineers Australia Stage 1 Competencies

This subject contributes to the development of the following Engineers Australia Stage 1 Competencies:

- 1.3. In-depth understanding of specialist bodies of knowledge within the engineering discipline.
- 1.5. Knowledge of engineering design practice and contextual factors impacting the engineering discipline.
- 2.1. Application of established engineering methods to complex engineering problem solving.
- 2.2. Fluent application of engineering techniques, tools and resources.
- 2.3. Application of systematic engineering synthesis and design processes.
- 2.4. Application of systematic approaches to the conduct and management of engineering projects.
- 3.2. Effective oral and written communication in professional and lay domains.
- 3.5. Orderly management of self, and professional conduct.
- 3.6. Effective team membership and team leadership.

UTS acknowledges the Gadigal people of the Eora Nation, the Boorooberongal people of the Dharug Nation, the Bidiagal people and the Gamaygal people, upon whose ancestral lands our university stands. We would also like to pay respect to the Elders both past and present, acknowledging them as the traditional custodians of knowledge for these lands.

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